

Byeongjee Kang

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Research Interests

My research is on language and compiler support for automatically generating safe and efficient programs. I combine static analysis with compiler infrastructure built on LLVM and MLIR. My recent interests are in resource-constrained platforms, where programs require checkpointing or migration to safely execute under strict energy and memory budgets.

Professional Skills

Programming Languages: Rust, C++, WebAssembly, Python, SML, Maude, Futhark

Frameworks and Tools: LLVM, MLIR, rustc (MIR), Gurobi

Embedded Systems: TI MSP430, Ambiq Apollo4 (Arm Cortex-M4F), bare-metal C, Otii Ace Pro (energy measurement), Saleae Logic (logic analyzer)

Education

PhD	Carnegie Mellon University , Software Engineering Advised by Prof. Limin Jia	Pittsburgh, USA Aug 2023 – May 2028 (expected)
BS	Pohang University of Science and Technology (POSTECH) , Computer Science and Engineering & Mathematics (double major) - GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023) - Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA - Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)	Pohang, Korea Feb 2018 – Feb 2023

Work Experiences

Microsoft Research , Research intern at Data Systems Group mentored by Kaushik Rajan Built auto-vectorizing compiler optimization passes using MLIR and Polygeist, vectorizing loops with control-flow dependencies that are not supported by existing loop vectorizers	Redmond, WA, USA May 2025 – Aug 2025
Kodebox , Software engineer as military service alternative - Developed the consensus engine of CodeChain Foundry, an open-source blockchain engine with a composable module system, implementing BLS signature aggregation and the interface between modules and the consensus engine - Built Zuzu, a web-based cap table management software, as a full-stack engineer	Seoul, Korea Jan 2020 – Dec 2021

Publications

Bao: Automatic Region Placement and Memory Allocation for Intermittent Computing

LLVM-based compiler that optimizes checkpoint placement and memory allocation for intermittently powered devices by solving a mixed-integer linear program

Byeongjee Kang, Jan Hoffmann, Limin Jia, Brandon Lucia, Feras Saad

(International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2027 (to appear)) [\[code\]](#)

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction

Aug 2026

MLIR-based compiler from high-level languages to WebAssembly that takes advantage of high-level Wasm features such as stack switching

Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia

[10.1145/3828170.3828172](https://doi.org/10.1145/3828170.3828172) (International Conference on Managed Programming Languages and Runtimes (MPLR) 2026) [\[code\]](#)

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM

Oct 2025

Ao Li, **Byeongjee Kang**, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye

[10.1145/3764119](https://doi.org/10.1145/3764119) (Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) 2025) [\[code\]](#)

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits

Sept 2024

Sam Westrick, Pengyu Liu, **Byeongjee Kang**, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar

ieeexplore.ieee.org/document/10821338 (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper))

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems

July 2024

Refutation-complete symbolic model checking of distributed systems in Maude using narrowing with heuristic search

Byeongjee Kang, Kyungmin Bae

[10.1016/j.scico.2024.103097](https://doi.org/10.1016/j.scico.2024.103097) (Science of Computer Programming 2024) [\[code\]](#)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search

Dec 2022

Byeongjee Kang, Kyungmin Bae

dl.acm.org/doi/10.1145/3563822.3568017 (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022) [\[code\]](#)

Software

- **Bao**: LLVM toolchain that instruments programs for batteryless devices by solving a mixed-integer linear program
- **WAMI**: MLIR-based compilation pipeline for WebAssembly
- **OCaMI** (work in progress): Compiler from a functional subset of OCaml to WebAssembly through WAMI
- **rustlane** (work in progress): ISPC-style SPMD programming for Rust, lowering scalar control flow to masked SIMD
- **Fray**: Controlled concurrency testing framework for the JVM
- **heuristic-narrowing-search**: Maude implementation of narrowing-based symbolic reachability analysis
- **Simperby**: BFT blockchain engine for decentralized autonomous organizations (DAOs). Designed **Vetomint**, its consensus protocol, which extends Tendermint with leader veto and early termination
- **CodeChain Foundry**: Open-source blockchain engine based on composable module system

Ongoing Research Projects

- **Language-Level Program Migration**: Compiler and runtime support for migrating live Rust programs under strict time and energy budgets by erasing reinitializable state at the source and rebuilding it at the destination
- **Probabilistic Energy Estimation**: Probabilistic modeling and analysis tools for estimating energy consumption of programs on resource-constrained devices
- **Cumulative Memory Analysis**: Type-based automatic amortized resource analysis that infers symbolic bounds on memory usage integrated over execution time

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation) Received total of USD 120,000 during graduate studies	Aug 2023 – July 2028
Global Leadership Program (CSE Department, POSTECH) Scholarship and research support program for selected students in the CS department	Sept 2019 – Feb 2023
Domun Scholarship (POSTECH) Scholarship for freshmen with the highest GPA	Feb 2019
Program for Highly Dedicated Students (POSTECH) Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year	Feb 2018 – Feb 2023
Presidential Science Scholarship (Ministry of Education of Korea) National scholarship offered to top freshmen students in the field of science and technology	Feb 2018 – Feb 2023

Talks

- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [\[recording\]](#)